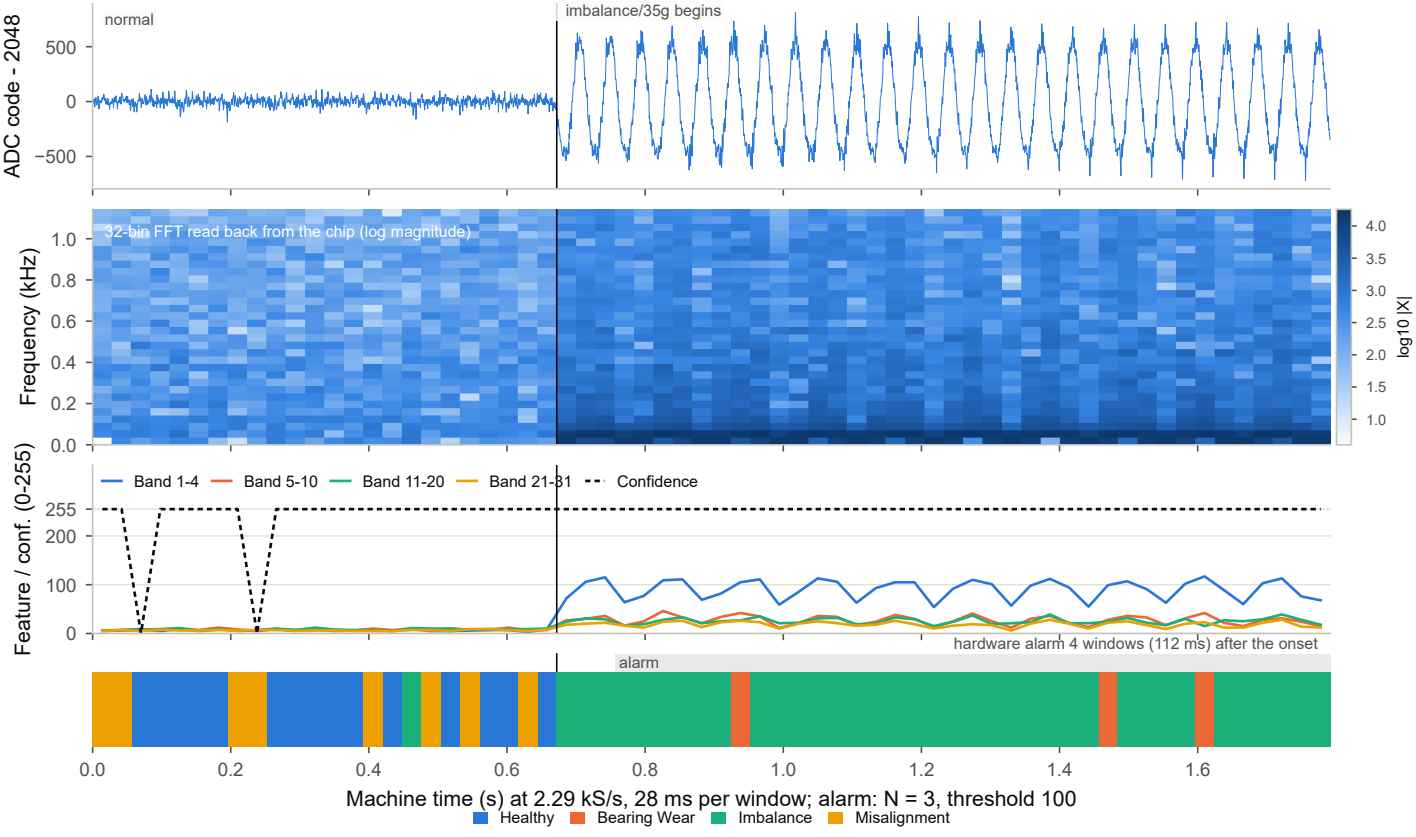

normal
imbalance/35g begins
ADC code - 2048
500
0
−500
32-bin FFT read back from the chip (log magnitude)
Frequency (kHz)
1.0
0.8
0.6
0.4
0.2
0.0
4.0
3.5
3.0
2.5
2.0
1.5
1.0
log10 |X|
Band 1-4
Band 5-10
Band 11-20
Band 21-31
Confidence
Feature / conf. (0-255)
255
200
100
0
hardware alarm 4 windows (112 ms) after the onset
alarm
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Machine time (s) at 2.29 kS/s, 28 ms per window; alarm: N = 3, threshold 100
Healthy
Bearing Wear
Imbalance
Misalignment